

HAFIZA LAIBA FAISAL

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PROFILE

Electrical Engineering undergraduate with demonstrated experience in aerospace systems, embedded systems, artificial intelligence, and power engineering. Recipient of multiple national awards including 1st Place in Pakistan's National CanSat Competition and Student of the Year. Experienced in developing real-world engineering solutions spanning healthcare, energy systems, disaster management, and intelligent autonomous platforms through research, industry, and open-source contributions.

EDUCATION

Bachelor of Electrical Engineering

Expected 2027

Lahore College for Women University (LCWU), Pakistan · CGPA: 3.22 / 4.00

Relevant Coursework: Signals & Systems · Control Systems · Power Electronics · Embedded Systems · Digital Logic Design · Probability Methods in Engineering

HONOURS & AWARDS

- 1st Position — Pakistan's First National CanSat Competition (SUPARCO), 2025
- Student of the Year — Lahore College for Women University
- Grid Girls Pakistan — Selected among top 12% nationwide (ADB-funded national initiative)
- Innovista AI Hackathon — Winner, FloodGuard AI
- Pakathon — 3rd Position (National Entrepreneurship Competition)
- International Recognition — NeuroShield, PeerBridge Mental Health Hacks

SELECTED TECHNICAL HIGHLIGHTS

- 3+ years of software and AI development experience across research, production systems, and competitions
- Production engineering experience at PenTutor (LLM-powered educational AI platform)
- National aerospace competition winner — real hardware deployment under competition conditions
- Industry experience in transmission system planning at NTDC (national grid operator)
- Open-source contributor: Blackwell OS (browser/desktop components) and MIPSDev (educational tooling)
- Led multidisciplinary engineering teams across aerospace, AI, embedded systems, and healthcare domains

INDUSTRY EXPERIENCE

AI Full Stack Software Engineer

Jan 2026 – Present

PenTutor — Production Educational AI Platform

- Architected LLM-powered student support systems and privacy-aware AI content moderation workflows serving real users.
- Designed role-based access control, scalable Django backend services, and deployment automation pipelines.

Power System Planning Intern

Jun 2025 – Aug 2025

National Transmission & Despatch Company (NTDC) — Competitive Selection, Top 12% Nationwide (Grid Girls Pakistan, ADB-funded)

- Conducted power flow studies and N-1 contingency analysis for Pakistan's national high-voltage transmission network using PSS®E.
- Performed dynamic stability analysis supporting long-range transmission planning and grid reliability assessments.

RESEARCH EXPERIENCE

EEG-Based Cognitive Load & Confusion Detection

Undergraduate Research

Lahore College for Women University

- Developed explainable ML models (XGBoost, SHAP, LIME) for cognitive state classification from EEG signals, targeting adaptive learning applications.
- Investigated physiological markers of human confusion and cognitive overload through signal processing and feature analysis.

SELECTED PROJECTS

SUPARCO National CanSat Mission System

1st Place — National Competition

- Developed and experimentally validated real-time 2D/3D trajectory reconstruction deployed on physical CanSat hardware during live national competition missions.
- Implemented full sensor fusion pipeline: accelerometer, gyroscope, magnetometer, GPS, and telemetry — with orientation estimation, geolocation, and impact prediction modules.

NeuroShield — EEG-Powered Addiction Recovery Platform

International Recognition, PeerBridge

- Developed an AI-assisted addiction recovery platform integrating real-time EEG signal analysis with intelligent behavioural support systems.

FloodGuard AI — Disaster Management Platform

Innovista Hackathon Winner

- Developed a multi-agent flood forecasting and evacuation planning system integrating satellite data, weather feeds, and GIS-based risk modelling.

CircuitSathi — AI Circuit Analysis & Debugging Assistant

Educational Technology

- Developed an AI-powered engineering assistant for circuit analysis, fault detection, debugging, and electrical engineering education.

HealthLink360 — Enterprise Healthcare Intelligence Platform

- Designed an AI-powered healthcare ecosystem integrating emergency response, pharmacy management, patient care coordination, and national health reporting.

Smart Energy Audit Monitoring System

Embedded Systems

- Developed an IoT energy monitoring platform using ESP32 and PZEM-004T; implemented real-time anomaly detection, remote control, and prepaid energy management.

OPEN SOURCE CONTRIBUTIONS

- Blackwell OS — Contributed browser components and desktop ecosystem features; participated in code review and collaborative release workflows.
- MIPSDev — Contributed validation testing and tooling improvements for a VS Code-based educational MIPS development extension.

LEADERSHIP & COMMUNITY ENGAGEMENT

Engineering Team Lead

Multiple Projects

- Led multidisciplinary teams across aerospace, AI, embedded systems, and healthcare domains from design through live competition deployment.
- Coordinated requirements scoping, technical implementation, testing, and formal presentations to faculty and competition judges.

AI Workshop Facilitator & Student Mentor

University Community

- Designed and facilitated Generative AI workshops for university students; mentored participants on applied AI, research pathways, and project development.
- Provided individual guidance on technical projects and career planning in AI and engineering research.

CERTIFICATIONS

- Huawei HCIA-AI — Huawei Certification Program
- Machine Learning Professional Certificate — IBM
- Multi AI Agent Systems with AutoGen — DeepLearning.AI
- AI Product Management — Microsoft
- Azure AI Apps & Agents — Microsoft Azure

TECHNICAL COMPETENCIES

Engineering & Systems: PSS®E · Embedded Systems (ESP32, Arduino) · Sensor Fusion · Telemetry Systems · ROS2 · Gazebo · MATLAB · Control Systems

AI & Machine Learning: Explainable AI (SHAP, LIME, XGBoost) · LLM Applications · Multi-Agent Systems · NLP · Computer Vision · EEG Signal Processing · RAG

Software Engineering: Python · C/C++ · JavaScript · Django · FastAPI · React · REST APIs · PostgreSQL · MongoDB · Linux

Domains: Healthcare Technology · Disaster Management · IoT & Energy Systems · GIS & Geospatial Analysis · Aerospace Engineering · Open Source Development

RESEARCH INTERESTS

Self-Healing Smart Grids · Quantum Technologies & Quantum Sensing · Molecular Communication & Nano-Scale Networks · Advanced Functional Materials (MOFs) for Energy Storage · Smart Sensing & Sustainable Infrastructure · Neuromorphic Computing & Brain-Inspired Hardware · Physics-Informed AI for Engineering Systems · Swarm Robotics · Embodied AI · Brain–Computer Interfaces (BCI) · Space Systems Engineering · AI for Scientific Discovery

References available upon request.